

Q: Are all of these enough to get full marks in the exam?

A: NO. This is a practice sheet. Meaning, you can practice all you want using the questions from this sheet. However, doing well in exams depends upon your ability to understand a question, formulate an answer, and express it correctly. You see, these are humane skills which cannot be guaranteed by completing a practice sheet only. But yeah, Best of luck anyway.

Chapter 2 (Instructions: Language of the Computer)

Question - 1:

Construct the equivalent RISC-V code of the following C code. Once you have the RISC-V code, identify type of each instruction and encode them accordingly.

$$\begin{aligned} A[7] &= A[2] + A[B[8]] + 10; \\ B[i] &= A[3] - 8; \end{aligned}$$

Base addresses of array A and B are in register X_{20} and X_{21} and i is in register X_{22}

Question - 2:

Construct the equivalent RISC-V code of the following C code.

```
for (i = 8; i > 0 ; i--) {  
    if ( A[i] == i){  
        A[2] = A [B[3]] ;  
    }  
}
```

Base addresses of array A and B are in register X_{20} and X_{21} . Also consider i is in register X_{22} .

Question - 3:

Construct the equivalent RISC-V code of the following C code.

```
if ( A[i] < i){  
    A[2] = A [B[3]] ;  
}
```

Base addresses of array A and B are in register X_{20} and X_{21} . Also consider i is in register X_{22} .

Question - 4:

Construct the equivalent RISC-V code of the following C code.

```
if ( A[3] != A[6]){  
    if (A[3] == 0) {  
        A[3] = A[3] + 2;  
    }else{  
        A[6] = A[6] / 16;  
    }  
}else{  
    A[6] = A[6] * 8  
}
```

Base addresses of array A and B are in register X_{20} and X_{21} .

Question - 5:

Construct the equivalent RISC-V code of the following C code.

```
Main () {  
    int x = 0;  
    int y = 9;  
    int z = addition(x, y);  
}
```

```

int addition (int a, int b) {
    int c = a + b;
    return c;
}

```

Variables x, y, z are stored in X_{20} , X_{21} and X_{22} registers. Argument x, y are passed using register X_{13} , X_{14}

Variable c from the addition function also uses register X_{21}

Question - 6:

Write RISC-V assembly code that checks if the number stored in register X_{25} is **even** or not. If **even** then store **1** in register X_{26} otherwise store **0**.

Question - 7:

ADD X_{25} , X_{25} , X_0 . Can you make this instruction faster? If yes, Write the updated instruction?

Question - 8:

Memory Location	Code	Line Number	Machine Code
	ADDI X_5 , X_0 , 5	1	
	ADDI X_6 , X_0 , 1	2	
	ADDI X_{25} , X_0 , 0	3	
	Loop: BLT X_5 , X_6 , loopBreak	4	_____XXX_____XXXXXXXX
	ADDI X_{25} , X_{25} , 1	5	
#7080	ADDI X_5 , X_5 , -1	6	
	BEQ X_0 , X_0 , Loop	7	_____XXX_____XXXXXXXX
	loopBreak:	8	

- What is the value of **PC** while executing line2? Answer: _____
- Fill up the machine codes corresponding to line4 and line7 in the table above.

Question - 9:

Memory Location	Code	Line Number
	Loop:	
	SLLI X_{10} , X_{22} , 3	1
	ADD X_{10} , X_{10} , X_{25}	2
	LD X_9 , 0(X_{10})	3
	BNE X_9 , X_{24} , Exit	4
#80016	ADDI X_{22} , X_{22} , 1	5
	BEQ X_0 , X_0 , Loop	6
	Exit:	

- Fill up the memory locations.
- Find the SB-type instructions from the above code and encode them accordingly.
Given,
I. opcode = $(103)_{10}$, funct3 = $(000)_2$ opcode for BEQ
II. opcode = $(103)_{10}$, funct3 = $(001)_2$ opcode for BNE

Question - 10:

Write necessary RISC-V instructions to store the value $(1111\ 1111\ 0000\ 1111\ 11)_2$ in X_{20} register.

Question - 11:

Show how the value 0xabcdef12 would be arranged in memory in RISC-V machine.

Question - 12:

For the RISC-V assembly instructions below, what is the corresponding C/high level statement?

slli x30, x5, 3 add x30, x10, x30 slli x31, x6, 3	Assume that the variables f, g, h, i, and j are assigned to registers x5, x6, x7,
---	---

<pre>add x31, x11, x31 ld x5, 0(x30) addi x12, x30, 8 ld x30, 0(x12) add x30, x30, x5 sd x30, 0(x31)</pre>	x28, and x29, respectively. Assume that the base address of the Arrays A and B are in registers x10 and x11, respectively.
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